

Experiment 9: k-Means Clustering

Title: Implementation of k-Means Clustering using Python

Objective: To understand and implement the k-Means clustering algorithm for unsupervised learning on a dataset.

Theory: k-Means is an unsupervised learning algorithm used for clustering data into k distinct groups based on similarity. The algorithm minimizes the sum of squared distances between each point and the centroid of its cluster.

The main idea is to define k centroids, one for each cluster, and assign each point to the nearest centroid, then update centroids by computing the mean of all points assigned to each cluster. This process is repeated until the centroids no longer change.

Algorithm

1. Initialize k cluster centers (centroids) randomly.
2. For each data point, calculate the **Euclidean distance** to all centroids.
3. Assign the data point to the nearest centroid based on the minimum distance.
4. Recalculate centroids as the mean of all points in the cluster.
5. Repeat steps 2 to 4 until convergence (i.e., when centroids no longer change or cluster assignments remain the same).

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.datasets import make_blobs

# Generate sample data
X, y_true = make_blobs(n_samples=300, centers=4, cluster_std=0.60,
random_state=0)

# Apply k-Means
kmeans = KMeans(n_clusters=4)
kmeans.fit(X)
y_kmeans = kmeans.predict(X)

# Plot the results
plt.scatter(X[:, 0], X[:, 1], c=y_kmeans, s=50, cmap='viridis')
centers = kmeans.cluster_centers_
```

```
plt.scatter(centers[:, 0], centers[:, 1], c='red', s=200, alpha=0.75, marker='X')  
plt.title("k-Means Clustering Result")  
plt.show()
```

Input

- Dataset: Randomly generated using `make_blobs`
- Parameter: Number of clusters $k = 4$

Output

- Cluster labels for each point
- Visualization of clusters and centroids

Analysis

k-Means efficiently clusters data by minimizing intra-cluster variance. However, it is sensitive to the initial centroid placement and the choice of k .

Conclusion

k-Means is a widely used clustering algorithm that works well with spherical clusters. The elbow method or silhouette score can help determine the best value for k .

Viva Questions

1. What is k-Means clustering?
2. Is k-Means supervised or unsupervised learning?
3. How is the number of clusters k chosen?
4. What are centroids in k-Means?
5. Explain the objective function in k-Means.
6. What are the limitations of k-Means?
7. How does k-Means handle outliers?
8. What distance metric is commonly used in k-Means?
9. Can k-Means be used with non-numeric data?
10. How do you determine if clustering is good?

Mathematical Questions

11. Given 3 data points $A(2,3)$, $B(6,7)$, $C(3,4)$, and initial centroids $C_1(2,2)$, $C_2(6,6)$, compute distances and assign clusters.
12. Define and explain the cost function used in k-Means.
13. Given a set of points, manually perform one iteration of the k-Means algorithm.
14. What is the formula for Euclidean distance in n -dimensional space?

15. Explain how the mean of cluster points is computed in each iteration.
16. What is the complexity of the k-Means algorithm?
17. Prove that the cost function in k-Means is non-increasing with each iteration.
18. What happens if two centroids are initialized close together?
19. Define inertia in the context of k-Means.
20. Given 4 points and $k = 2$, manually calculate new centroids after initial assignment.